

Bank Term Deposit Subscription Prediction Using Machine Learning

Israt Zaman Srity 2211084042

Group 12

August 7, 2026

Literature Review

Dong et al. [1] used machine learning techniques to predict whether bank customers would subscribe to a term deposit using the Portuguese Bank Marketing dataset. The dataset contained 41,188 customer contacts, with only 11.3% positive subscription rate, creating a severe class imbalance. To address this issue, the authors combined synthetic minority oversampling (SMOTE) with cost-sensitive learning across eight machine learning models, including Logistic Regression, Decision Tree, Random Forest, and ensemble methods, evaluated under five-fold stratified cross-validation. Ensemble models, particularly CatBoost, XGBoost, and LightGBM, outperformed traditional baselines. The best-performing model achieved an F1 Score of 0.540, a recall of 0.812 for the positive class, and a ROC-AUC of 0.908. SHAP-based analysis was further applied to identify campaign duration, previous contact outcomes, and macroeconomic indicators as the most influential predictors.

References

- [1] B. Dong, X. Zhang, Y. Liu, T. Zhang, X. Liu, L. Hou, L. Meng, Z. Guo, and A. Mulati, “Risk-Sensitive Machine Learning for Financial Decision Modeling Under Imbalanced Data: Evidence from Bank Telemarketing,” *Entropy*, vol. 28, pp. 354, 2026.